



National University

of Computer & Emerging Sciences-Islamabad
Chiniot-Faisalabad Campus

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AI 2002: Artificial Intelligence

Assignment No. 3

Due Date: Friday, 04 April 2024 (11:55 pm)

Instructions:

1. Submit your assignment in **soft form (Code + Report)** within the due date and time. Soft form does not mean submitting photos of the hardcopy.
 2. The **report** must include a discussion, comments, and a conclusion about your solution. Submitting without a report will result in a loss of marks.
 3. Name the zip or other folder/file that you submit using the following format: **AI_A1_RollNo_FirstName**.
 4. Ensure that you solve each task of the assignment on your own.
 5. Late submissions will result in a deduction of marks.
 6. For programming tasks, you are NOT allowed to use any built-in functions or libraries for specific tasks.
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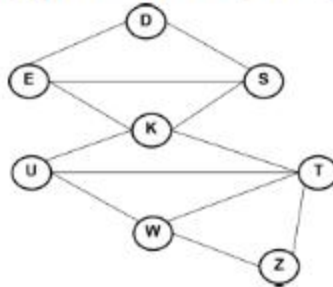
Question No. 1: [CSP]

- (a) Consider the following logic puzzle: In five houses, each with a different color, live five persons of different nationalities, each of whom prefers a different brand of candy, a different drink, and a different pet. Given the following facts, the questions to answer are “Where does the zebra live, and in which house do they drink water?”

- The Englishman lives in the red house.
- The Spaniard owns the dog.
- The Norwegian lives in the first house on the left.
- The green house is immediately to the right of the ivory house.
- The man who eats Hershey bars lives in the house next to the man with the fox.
- Kit Kats are eaten in the yellow house.
- The Norwegian lives next to the blue house.
- The Smarties eater owns snails.
- The Snickers eater drinks orange juice.

- The Ukrainian drinks tea.
- The Japanese eats Milky Ways.
- Kit Kats are eaten in a house next to the house where the horse is kept.
- Coffee is drunk in the green house.
- Milk is drunk in the middle house.

(b) Implement the constraint satisfaction problem using **Forward checking with backtracking** of the following map coloring problem. The state space is represented as,



Variables: D, S, E, K, U, T, W, Z

Domains: $D_i = \{\text{red; green; blue}\}$

Constraints: adjacent regions must have different colours.

Question No. 2 [AC-3]

Implement the AC-3 algorithm to show that arc consistency can detect the inconsistency of the partial assignment $\{WA=\text{green}, V=\text{red}\}$ for the problem shown in Figure 1.

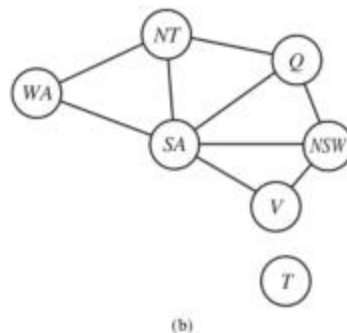


Figure 1: (a) The principal states and territories of Australia. Coloring this map can be viewed as a constraint satisfaction problem (CSP). The goal is to assign colors to each region so that no neighboring regions have the same color. **(b)** The map-coloring problem represented as a constraint graph.

Question No. 3: [Entailment and Sentence Validity]

(a) Prove each of the following assertions:

- I. α is valid if and only if $\text{True} \models \alpha$.
- II. For any α , $\text{False} \models \alpha$.
- III. $\alpha \models \beta$ if and only if the sentence $(\alpha \Rightarrow \beta)$ is valid.
- IV. $\alpha = \beta$ if and only if the sentence $(\alpha \Leftrightarrow \beta)$ is valid.
- V. $\alpha \models \beta$ if and only if the sentence $(\alpha \wedge \neg \beta)$ is unsatisfiable.

(b) Prove, or find a counterexample to, each of the following assertions:

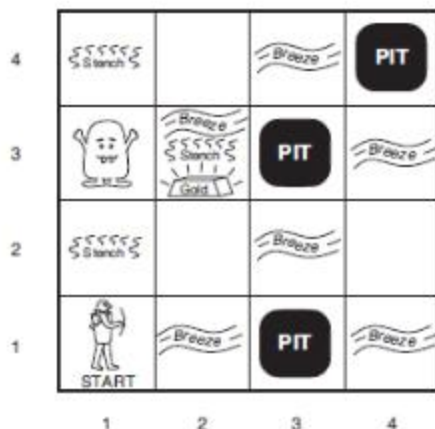
- I. If $\alpha \models \gamma$ or $\beta \models \gamma$ (or both) then $(\alpha \wedge \beta) \models \gamma$.
- II. If $\alpha \models (\beta \wedge \gamma)$ then $\alpha \models \beta$ and $\alpha \models \gamma$.
- III. If $\alpha \models (\beta \vee \gamma)$ then $\alpha \models \beta$ or $\alpha \models \gamma$ (or both).

Question No. 4: [Inference in Propositional Logic]

[Wumpus world]: Consider the Wumpus world example with the following rules:

- ▶ The agent can move *Forward*, *TurnLeft* by 90° , *TurnRight* by 90° , grab, shoot
- ▶ The square adjacent directly (not diagonally) to the square containing **Wumpus**, the agent will perceive a *Stench*.
- ▶ The squares adjacent to a **pit**, the agent will perceive a *Breeze*.
- ▶ The square with **gold**, the agent will perceive a *Glitter*.
- ▶ An agent **walks into a wall**, it will perceive a *Bump*.
- ▶ When the **Wumpus is killed**, it emits a woeful *Scream*.

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2	3,2	4,2
OK			
1,1	2,1	3,1	4,1
A			
OK	OK		



The given knowledge base is,

$$R_1: \neg P_{1,1} .$$

$$R_2: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}) .$$

$$R_3: B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1}) .$$

$$R_4: \neg B_{1,1} .$$

$$R_5: B_{2,1} .$$

$$R_6: S_{1,2} .$$

$$R_7: \neg S_{1,1} .$$

$$R_8: S_{1,2} \Leftrightarrow (W_{1,3} \vee W_{2,2}) .$$

$P_{1,1}$ = Pit in [1,1],

$B_{1,1}$ = Breeze in [1,1]

$S_{1,1}$ = Stench in [1,1],

$W_{1,1}$ = Wumpus in [1,1],

- Apply the inference rules and derive the proof step by step that $\neg P_{1,2}$.
- Apply the inference rules and derive the proof step by step that $P_{3,1}$.
- Apply the inference rules and derive the proof step by step that $W_{1,3}$.
- Apply the resolution theorem and derive the proof step by step that $\neg P_{1,2}$.
- Apply the resolution theorem and derive the proof step by step that $W_{1,3}$.

Note: If any of the above five statements cannot be proven, provide a valid and suitable justification.